Q.P Code D 122646	Total Pages 2	Name 601740
		Register No.

SECOND SEMESTER (CUFYUGP) DEGREE EXAMINATION, APRIL 2025

MATHEMATICS

MAT2CJ101-INTEGRAL CALCULUS

2024 Admission Onwards

Maximum Time :2 Hours	**Maximum Marks :70**

Section A

All Question can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Evaluate $\int \sin^2 x dx$
2	Find $\int \cos(7\theta + 5)\ d\theta$
3	Graph the integrand and use areas to evaluate the integral $\int_2^5 (2x + 1)\ dx$.
4	State Fundamental Theorem of Calculus
5	Find $\int \tan x\ dx$
6	Find y if $\ln y = t/2 + 7$
7	Find $\lim_{x \to 0} \frac{x^2}{\sin x}$
8	Express $\frac{5x - 3}{x^2 - 2x - 3}$ as a sum of partial fractions.
9	Explain the method to find the area between two curves
10	Set up an integral for the length of the curve $y = x^2$; $-1 \leq x \leq 2$

Section B

All Question can be answered. Each Question carries 6 marks (Ceiling : 36 Marks))

11	Solve $\int \sqrt{4t-1}\, dt$
12	Suppose that f is continuous and that $\int_1^2 f(x)dx = 4$. Show that $f(x) = 4$ at least once on $[1, 2]$.
13	Evaluate the integral $\int_0^{\pi} \frac{\sec x \tan x}{2 + \sec x}\, dx$
14	Solve the initial value problem $e^y \frac{dy}{dx} = 2x,\ x > \sqrt{3}, y(2) = 0$
15	Evaluate $\lim_{x \to 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$
16	Evaluate $\int \frac{1}{\sqrt{4x - x^2}}\, dx$
17	Find the areas of the regions enclosed by the curves $x + y^2 = 3$ and $4x + y^2 = 0$
18	Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$.

Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks))

19	Graph the function $f(x) = x^2 - 6x + 8$ on the interval $[0, 3]$. Then find 1. find the area of the region between the graph and the x -axis. 2. average value of $f(x)$ over that interval
20	Find the length of the curve $x = \frac{y^3}{3} + \frac{1}{4y}$ from $y = 1$ to $y = 3$